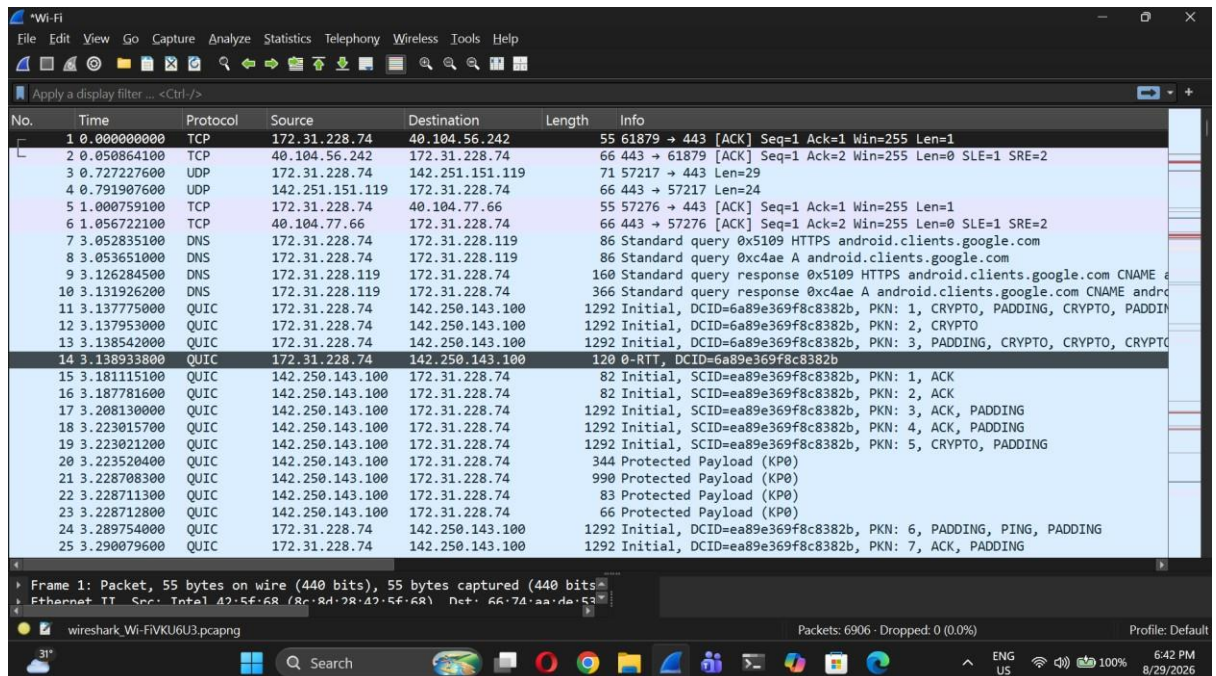


CNAM ASSIGNMENT

1. List up to 10 different protocols that appear in the Protocol column in the unfiltered packet-listing window.

ANS : 1



No.	Time	Protocol	Source	Destination	Length	Info
1	0.000000000	TCP	172.31.228.74	40.104.56.242	55	61879 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
2	0.050864100	TCP	40.104.56.242	172.31.228.74	66	443 → 61879 [ACK] Seq=1 Ack=2 Win=255 Len=0 SLE=1 SRE=2
3	0.727227600	UDP	172.31.228.74	142.251.151.119	71	57217 → 443 Len=29
4	0.791907600	UDP	142.251.151.119	172.31.228.74	66	443 → 57217 Len=24
5	1.000759100	TCP	172.31.228.74	40.104.77.66	55	57276 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6	1.056722100	TCP	40.104.77.66	172.31.228.74	66	443 → 57276 [ACK] Seq=1 Ack=2 Win=255 Len=0 SLE=1 SRE=2
7	3.052835100	DNS	172.31.228.74	172.31.228.119	86	Standard query 0x5109 HTTPS android.clients.google.com
8	3.053651000	DNS	172.31.228.74	172.31.228.119	86	Standard query 0xc4ae A android.clients.google.com
9	3.126284500	DNS	172.31.228.119	172.31.228.74	160	Standard query response 0x5109 HTTPS android.clients.google.com CNAME
10	3.131926200	DNS	172.31.228.119	172.31.228.74	366	Standard query response 0xc4ae A android.clients.google.com CNAME andr
11	3.137750000	QUIC	172.31.228.74	142.250.143.100	1292	Initial, DCID=6a89e369f8c8382b, PKN: 1, CRYPTO, PADDING, CRYPTO, PADDIN
12	3.137953000	QUIC	172.31.228.74	142.250.143.100	1292	Initial, DCID=6a89e369f8c8382b, PKN: 2, CRYPTO
13	3.138542000	QUIC	172.31.228.74	142.250.143.100	1292	Initial, DCID=6a89e369f8c8382b, PKN: 3, PADDING, CRYPTO, CRYPTO, CRYPTO
14	3.138933800	QUIC	172.31.228.74	142.250.143.100	120	0-RTT, DCID=6a89e369f8c8382b
15	3.181115100	QUIC	142.250.143.100	172.31.228.74	82	Initial, SCID=ea89e369f8c8382b, PKN: 1, ACK
16	3.187781600	QUIC	142.250.143.100	172.31.228.74	82	Initial, SCID=ea89e369f8c8382b, PKN: 2, ACK
17	3.208130000	QUIC	142.250.143.100	172.31.228.74	1292	Initial, SCID=ea89e369f8c8382b, PKN: 3, ACK, PADDING
18	3.223015700	QUIC	142.250.143.100	172.31.228.74	1292	Initial, SCID=ea89e369f8c8382b, PKN: 4, ACK, PADDING
19	3.223021200	QUIC	142.250.143.100	172.31.228.74	1292	Initial, SCID=ea89e369f8c8382b, PKN: 5, CRYPTO, PADDING
20	3.223520400	QUIC	142.250.143.100	172.31.228.74	344	Protected Payload (KP0)
21	3.228708300	QUIC	142.250.143.100	172.31.228.74	990	Protected Payload (KP0)
22	3.228711300	QUIC	142.250.143.100	172.31.228.74	83	Protected Payload (KP0)
23	3.228712800	QUIC	142.250.143.100	172.31.228.74	66	Protected Payload (KP0)
24	3.289754000	QUIC	172.31.228.74	142.250.143.100	1292	Initial, DCID=ea89e369f8c8382b, PKN: 6, PADDING, PING, PADDING
25	3.290079600	QUIC	172.31.228.74	142.250.143.100	1292	Initial, DCID=ea89e369f8c8382b, PKN: 7, ACK, PADDING

2. What is the Internet address of the www.gujaratuniversity.ac.in?
What is the Internet address of your computer?
->nslookup www.gujaratuniversity.ac.in
->ipconfig

ANS : 2

MY IP ADDRESSES :

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::4399:6ed2:233e:bad3%14
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-Local IPv6 Address . . . . . : fe80::69a4:e6ce:4250:e1af%10
    IPv4 Address. . . . . : 172.31.228.74
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.31.228.119

C:\Users\Dell>
```

IP ADDRESSED OF www.gujrtuniversity.ac.in :

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>nslookup www.gujaratuniversity.ac.in
Server:      Unknown
Address:     172.31.228.119

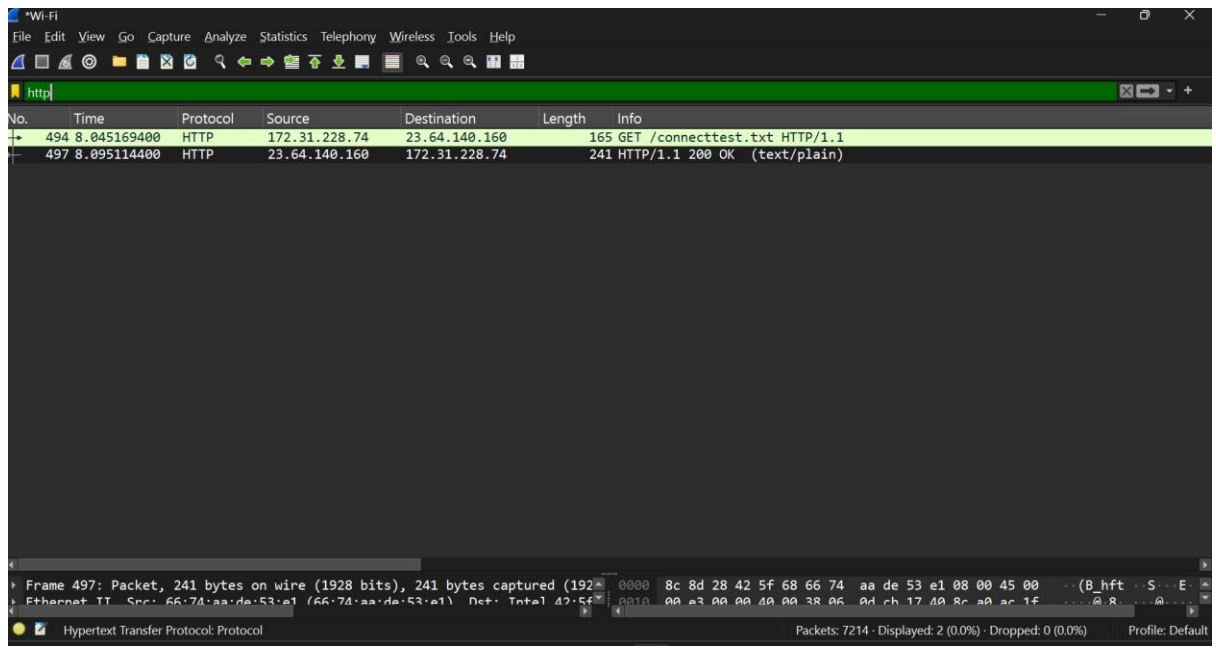
Non-authoritative answer:
Name:   gujaratuniversity.ac.in
Addresses:  64:ff9b::e8b:7a12
           14.139.122.18
Aliases:  www.gujaratuniversity.ac.in

C:\Users\Dell>
```

3. http protocol only with Wireshark running on your computer.

ANS : 3

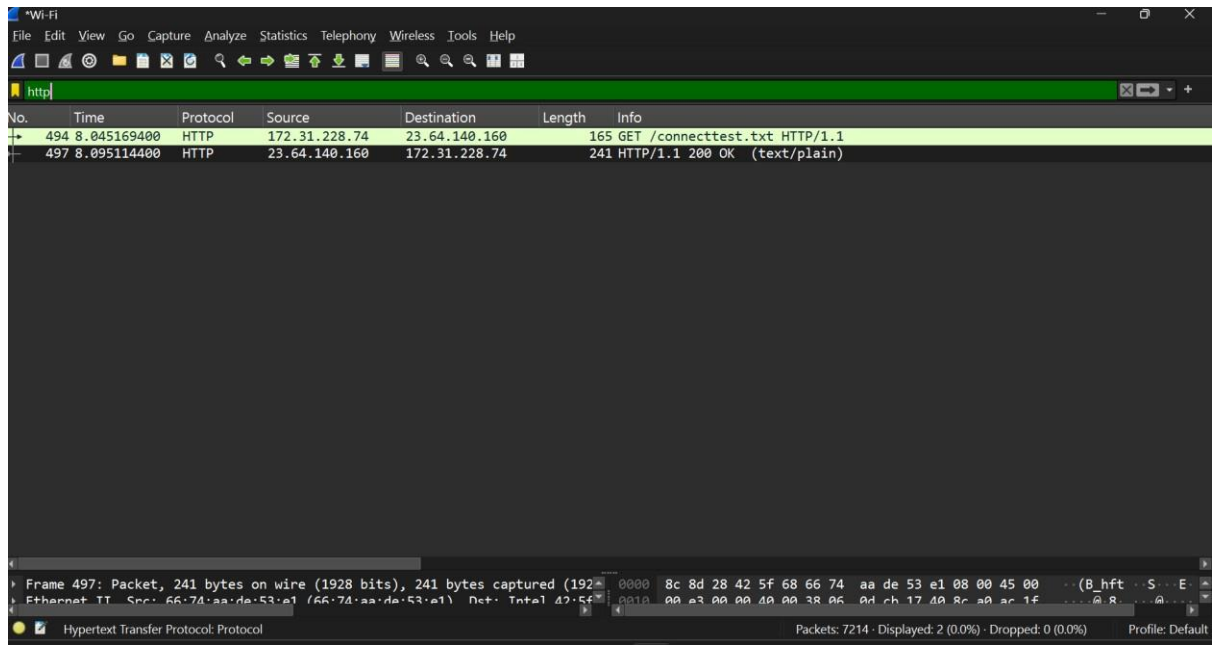
Total HTTP packets : 2



4. How many packets did you capture (total of all protocols, not just HTTP)?

ANS : 4

Total packets : 7214



5. Now, use display filters to determine how many packets contain your ip address (hint: Use ip.addr instead of the clumsy ip.src or ip.dst format). ANS : 5

My IP address = 172.31.228.74 IP address:7206 packets

No.	Time	Protocol	Source	Destination	Length	Info
485	8.007937300	DNS	172.31.228.74	172.31.228.119	83	Standard query 0x56a6 A www.msftconnecttest.com
486	8.010730600	TCP	172.31.228.74	23.64.140.160	66	63556 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
487	8.015230500	DNS	172.31.228.119	172.31.228.74	196	Standard query response 0x56a6 A www.msftconnecttest.com CNAME www.msft
488	8.034772600	QUIC	142.251.157.119	172.31.228.74	1210	Protected Payload (KP0)
489	8.034775300	QUIC	142.251.157.119	172.31.228.74	68	Protected Payload (KP0)
490	8.034776400	QUIC	142.251.157.119	172.31.228.74	159	Protected Payload (KP0)
491	8.035584500	QUIC	172.31.228.74	142.251.157.119	78	Protected Payload (KP0), DCID=e12d3f95a9b6aa83
492	8.043334200	TCP	23.64.140.160	172.31.228.74	66	80 → 63556 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1300 SACK_PERM WS
493	8.043607900	TCP	172.31.228.74	23.64.140.160	54	63556 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
494	8.045169400	HTTP	172.31.228.74	23.64.140.160	165	GET /connecttest.txt HTTP/1.1
495	8.063399700	QUIC	172.31.228.74	142.251.157.119	75	Protected Payload (KP0), DCID=e12d3f95a9b6aa83
496	8.073505000	TCP	23.64.140.160	172.31.228.74	54	80 → 63556 [ACK] Seq=1 Ack=112 Win=64256 Len=0
497	8.095114400	HTTP	23.64.140.160	172.31.228.74	241	HTTP/1.1 200 OK (text/plain)
498	8.095115700	TCP	23.64.140.160	172.31.228.74	54	80 → 63556 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
499	8.096789400	TCP	172.31.228.74	23.64.140.160	54	63556 → 80 [ACK] Seq=112 Ack=189 Win=65280 Len=0
500	8.097070700	TCP	172.31.228.74	23.64.140.160	54	63556 → 80 [FIN, ACK] Seq=112 Ack=189 Win=65280 Len=0
501	8.097126300	TCP	172.31.228.74	23.64.140.160	54	63556 → 80 [RST, ACK] Seq=113 Ack=189 Win=0 Len=0
502	8.102620800	QUIC	142.251.157.119	172.31.228.74	68	Protected Payload (KP0)
503	8.127700100	TCP	23.64.140.160	172.31.228.74	54	[TCP Retransmission] 80 → 63556 [FIN, ACK] Seq=188 Ack=112 Win=64256 L
504	8.127815000	TCP	172.31.228.74	23.64.140.160	54	63556 → 80 [RST] Seq=112 Win=0 Len=0
505	8.450619500	QUIC	172.31.228.74	142.251.157.119	259	Protected Payload (KP0), DCID=e12d3f95a9b6aa83
506	8.515702000	QUIC	142.251.157.119	172.31.228.74	72	Protected Payload (KP0)
507	8.542698700	QUIC	172.31.228.74	142.251.157.119	75	Protected Payload (KP0), DCID=e12d3f95a9b6aa83
508	8.588085900	QUIC	142.251.157.119	172.31.228.74	1173	Protected Payload (KP0)
509	8.588087700	QUIC	142.251.157.119	172.31.228.74	68	Protected Payload (KP0)

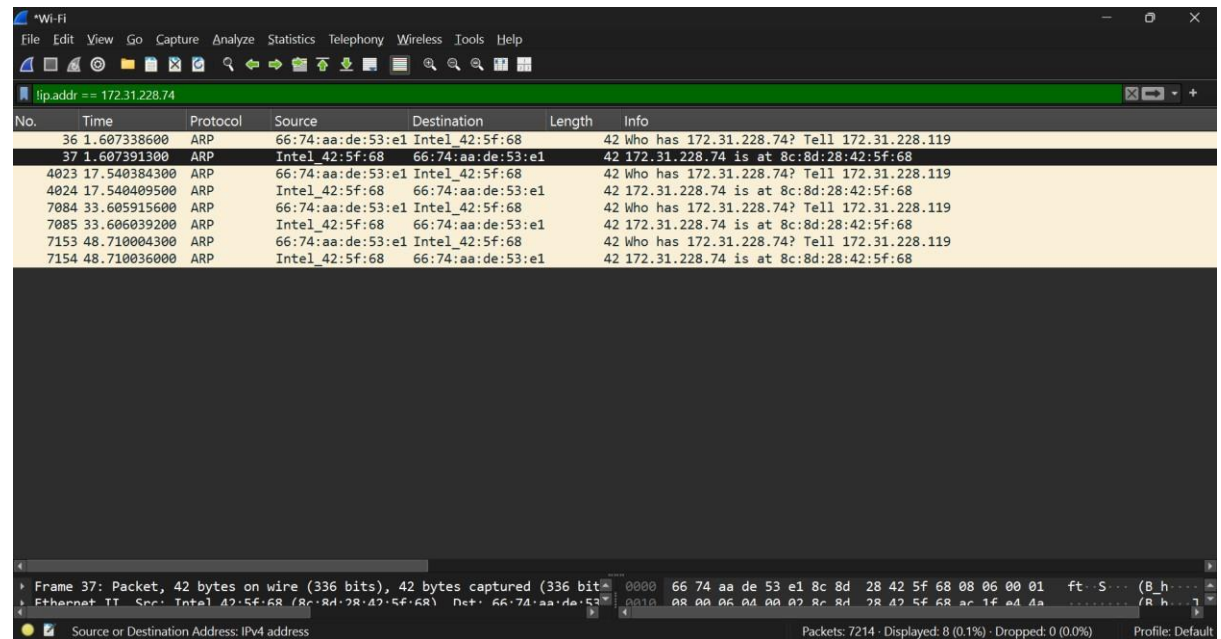
Frame 497: Packet, 241 bytes on wire (1928 bits), 241 bytes captured (1928 bits) on interface 0
 Ethernet II Src: 66:7d:aa:de:53:a1 (66:7d:aa:de:53:a1) Dst: Intel 47:56:01:00:00:00
 Source or Destination Address: IPv4 address

Packets: 7214 · Displayed: 7206 (99.9%) · Dropped: 0 (0.0%) · Profile: Default

6. What is this filter you used? Now, reverse the filter to determine how many packets don't contain your ip address.

ANS : 6

Not content IP addresses : 8 pockets



The image shows a Wireshark packet capture window titled "Wi-Fi". The filter bar at the top shows "ip.addr == 172.31.228.74". The packet list shows several ARP packets. The selected packet (No. 37) is an ARP request from Intel_42:5f:68 to 66:74:aa:de:53:e1. The packet details pane shows the Ethernet II header and the ARP section. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Protocol	Source	Destination	Length	Info
36	1.607338600	ARP	66:74:aa:de:53:e1	Intel_42:5f:68	42	Who has 172.31.228.74? Tell 172.31.228.119
37	1.607391300	ARP	Intel_42:5f:68	66:74:aa:de:53:e1	42	172.31.228.74 is at 8c:8d:28:42:5f:68
4023	17.540384300	ARP	66:74:aa:de:53:e1	Intel_42:5f:68	42	Who has 172.31.228.74? Tell 172.31.228.119
4024	17.540409500	ARP	Intel_42:5f:68	66:74:aa:de:53:e1	42	172.31.228.74 is at 8c:8d:28:42:5f:68
7084	33.605915600	ARP	66:74:aa:de:53:e1	Intel_42:5f:68	42	Who has 172.31.228.74? Tell 172.31.228.119
7085	33.606039200	ARP	Intel_42:5f:68	66:74:aa:de:53:e1	42	172.31.228.74 is at 8c:8d:28:42:5f:68
7153	48.710004300	ARP	66:74:aa:de:53:e1	Intel_42:5f:68	42	Who has 172.31.228.74? Tell 172.31.228.119
7154	48.710036000	ARP	Intel_42:5f:68	66:74:aa:de:53:e1	42	172.31.228.74 is at 8c:8d:28:42:5f:68

Frame 37: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
Ethernet II, Src: Intel_42:5f:68 (8c:8d:28:42:5f:68), Dst: 66:74:aa:de:53:e1 (66:74:aa:de:53:e1)
ARP, Operation: request, Sender MAC: 66:74:aa:de:53:e1, Target MAC: 8c:8d:28:42:5f:68, Sender IP: 172.31.228.119, Target IP: 172.31.228.74

7. What is the IP address of your computer? Of the gtu.ac.in server?
(nslookup gtu.ac.in)

ANS : 7

IP address of my computer :

```

C:\Users\Dell>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::4399:6ed2:233e:bad3%14
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::69a4:e6ce:4250:e1af%10
    IPv4 Address. . . . . : 172.31.228.74
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.31.228.119

C:\Users\Dell>

```

IP address of gtu.ac.in :

```

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>nslookup www.gtu.ac.in
Server:      UnKnown
Address:     172.31.228.119

Non-authoritative answer:
Name:       www.gtu.ac.in
Addresses:  64:ff9b::d7e:9aa6
            64:ff9b::dcf:9f94
            13.126.154.166
            13.207.159.148

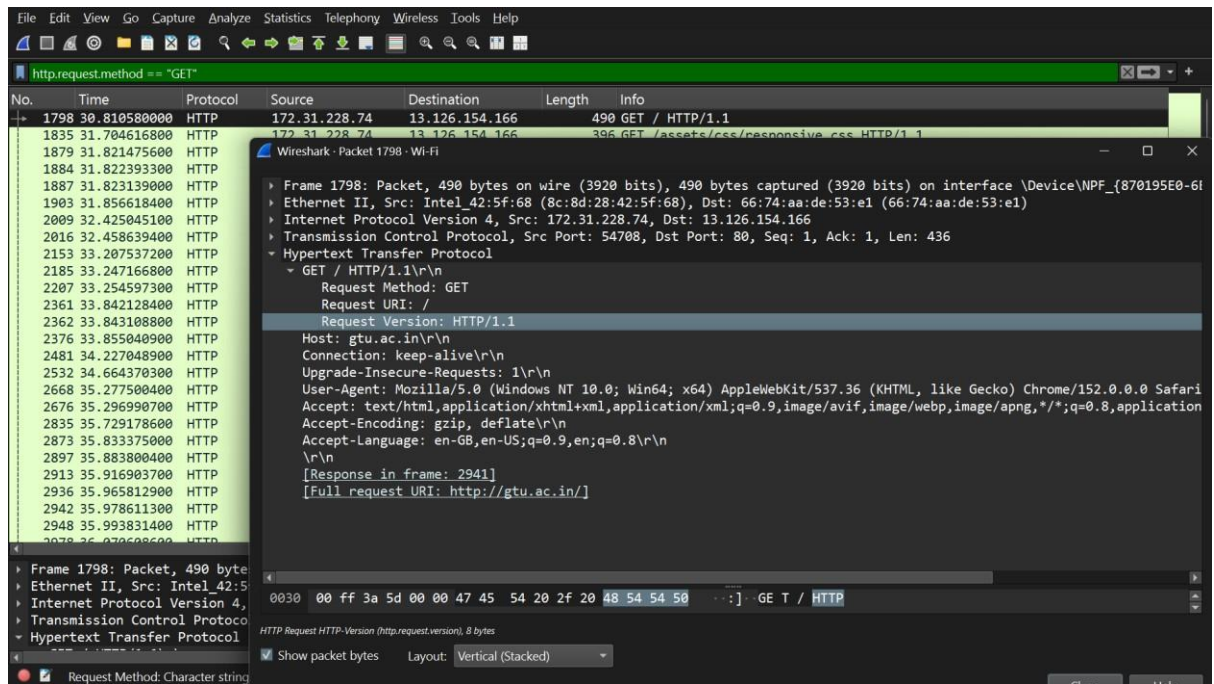
C:\Users\Dell>

```

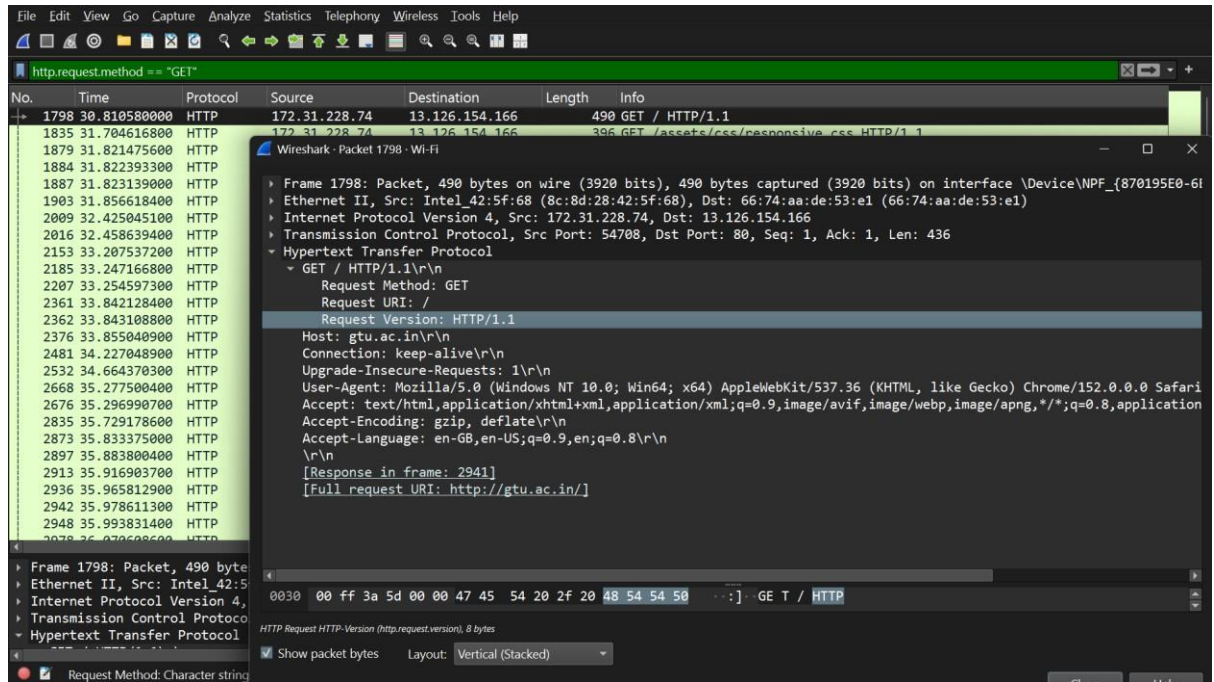
8. What HTTP version is your browser running? What version of HTTP is the server running?

ANS : 8

Request version : 1.1



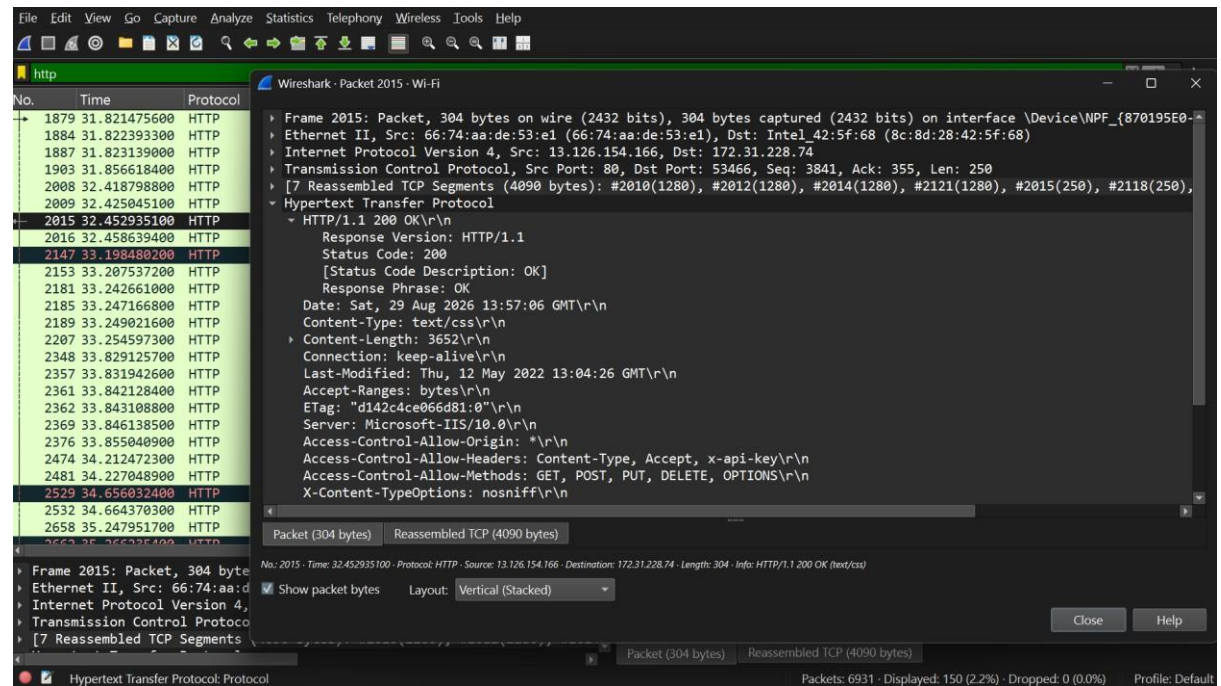
Response version 1.1



9. What is the status code and phrase returned from the server to your browser?(http.response.code)

ANS : 9

Status code 200 Response phrase : OK

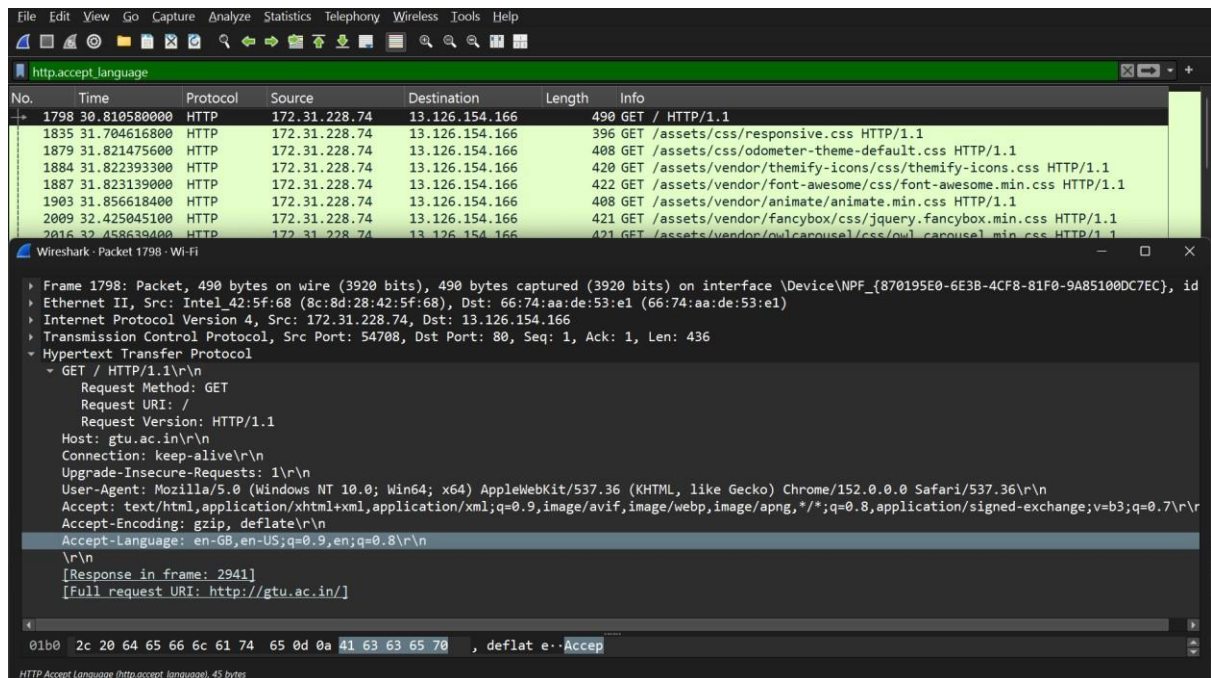


10.What languages does your browser indicate to the server that it can accept? Which header line is used to indicate this information?

http.request or http.accept_language

ANS : 10

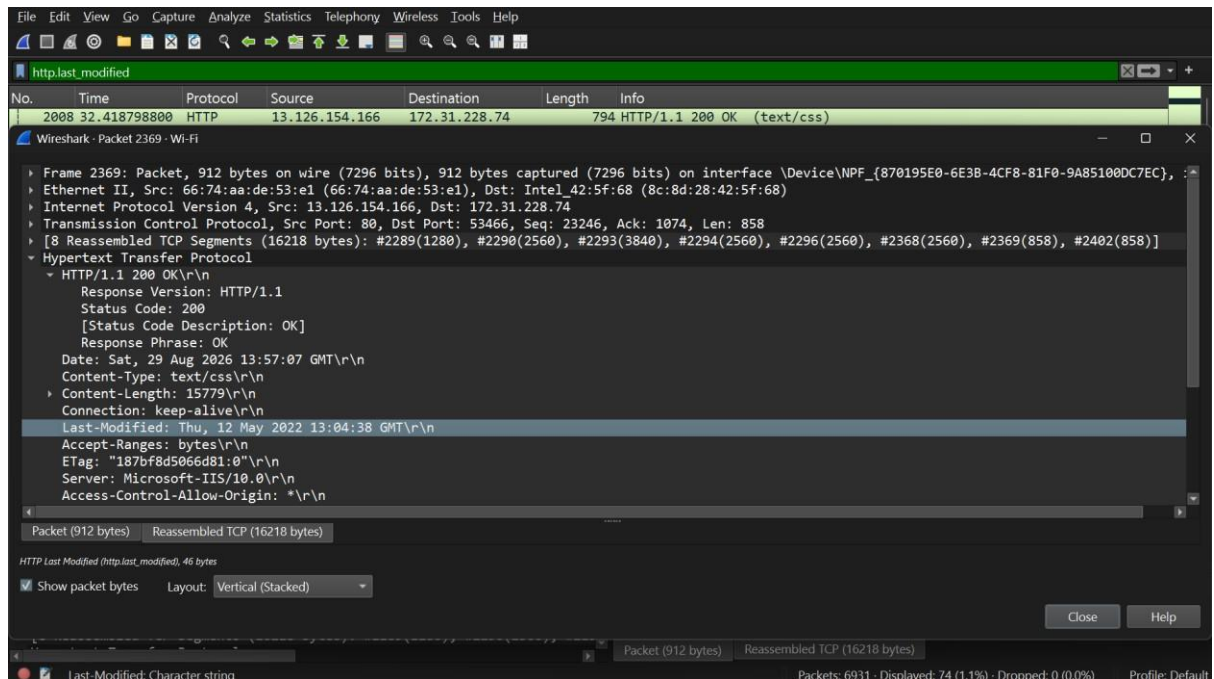
Accept language : en-GB , en-US ; q=0.9,en;q=0.8\r\n



11. When was the HTML-file, that you have retrieved, last modified at the server? Which header line is used to indicate this information? (When was this webpage last updated on the server) `http.response.code == 200 && http.last_modified`

ANS : 11

Last modified : Thu,12 May 2022 13:04:38



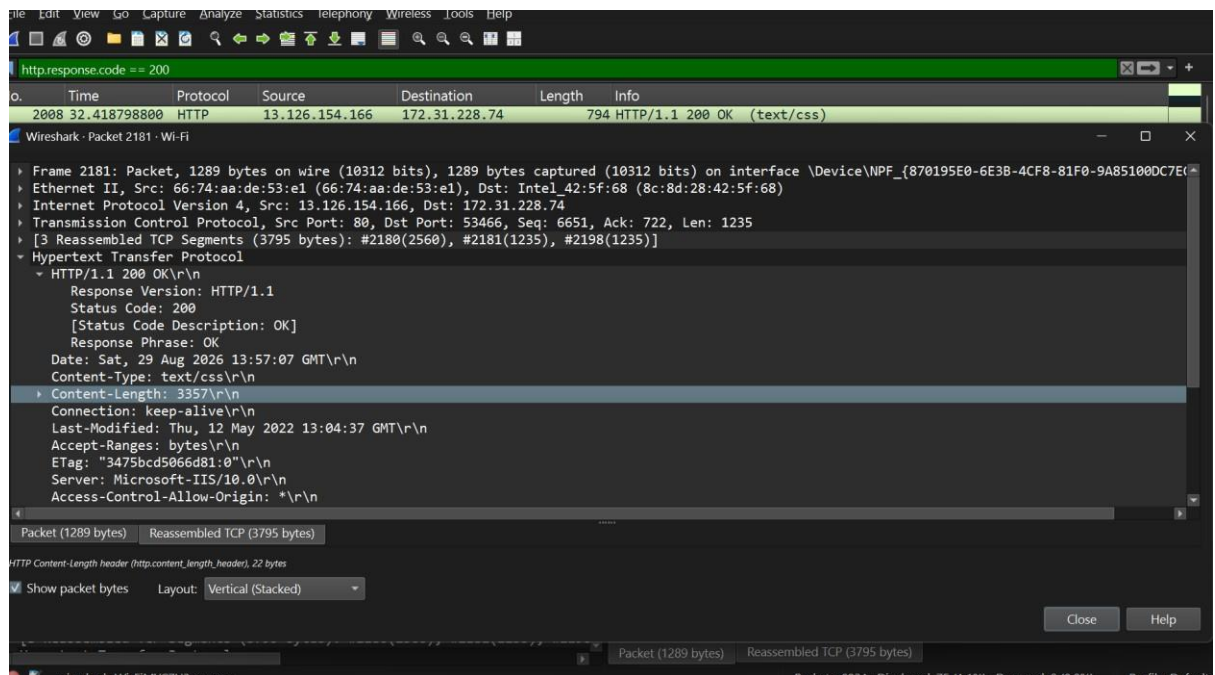
12.How many bytes of content (size of file) are returned to your browser?

Which header line is used to indicate this information?

(When the server sends the webpage (HTML file) to your browser, what is the size of that file?)

ANS : 12

Content-length : 3357\r\n



13. Start up the Wireshark packet sniffer, and make sure that “http” is in the display filter, so that only captured HTTP messages will be displayed in the packet-list pane.

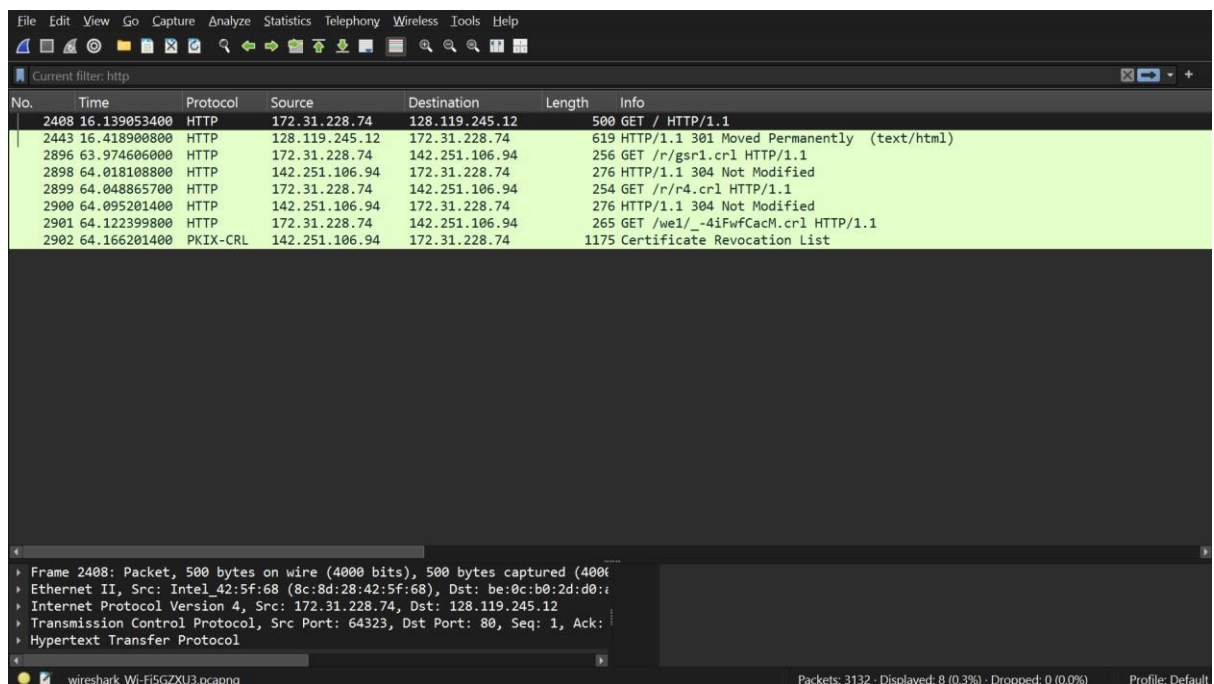
14. Enter the following URL into your browser:

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html>

Your browser should display a very simple HTML file.

15. Quickly enter the same URL into your browser again (or simply select the refresh button on your browser).

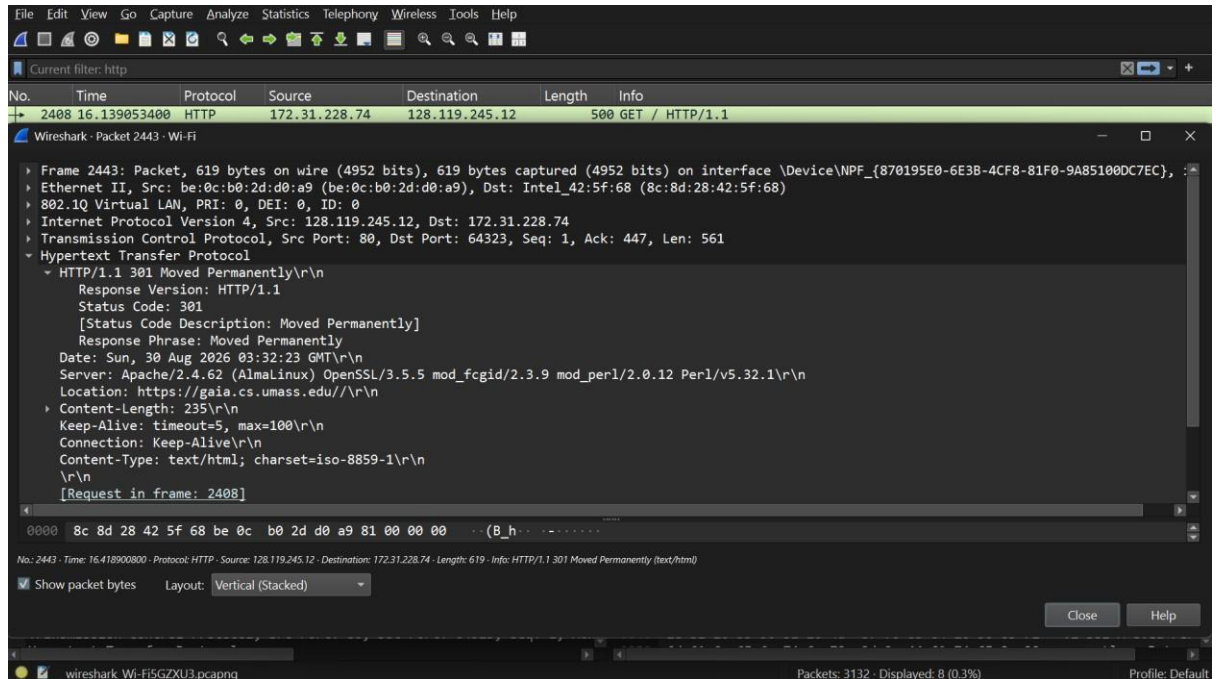
16. Stop Wireshark packet capture.



17. Inspect the contents of the first HTTP GET request from your browser to the server. Is there an “IF-MODIFIED-SINCE” header line in the HTTP GET message? Why or why not? (1st link - no)

Ans: 17

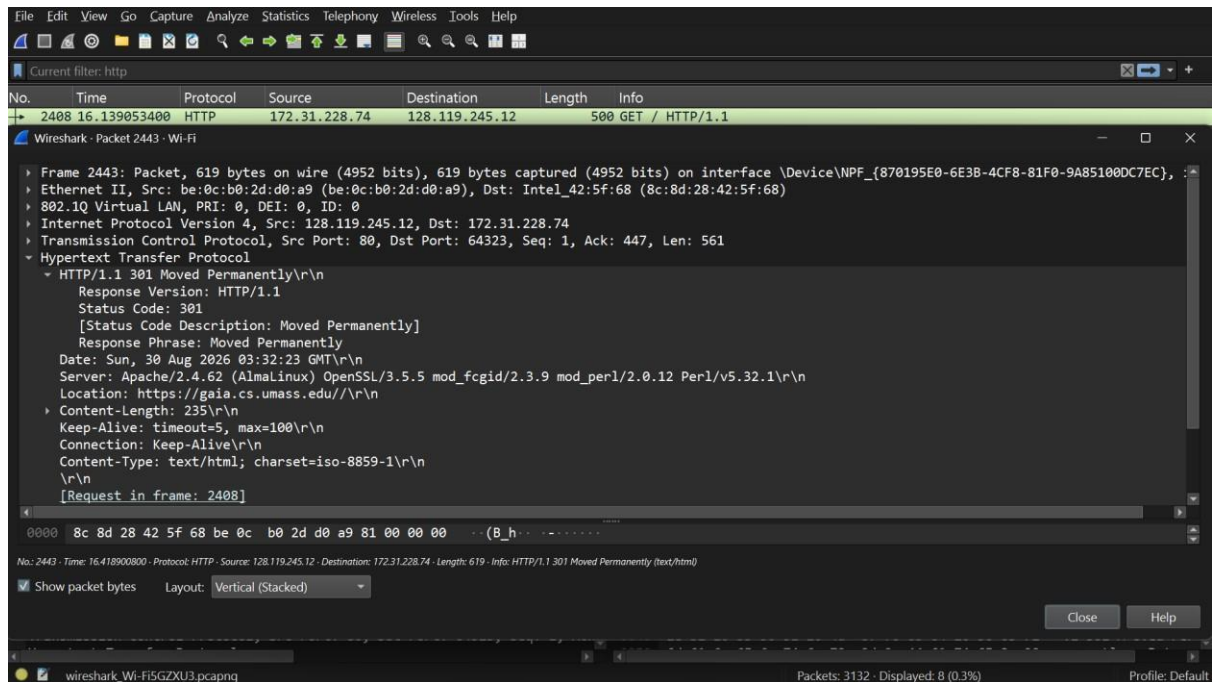
No. If-Modified-Since is not present because the browser is requesting the page for the first time. It has no previous copy of the page, so it simply asks the server to send the page.



18. Inspect the contents of the server response. Has the server explicitly returned the contents of the file? How can you tell?

Ans :18

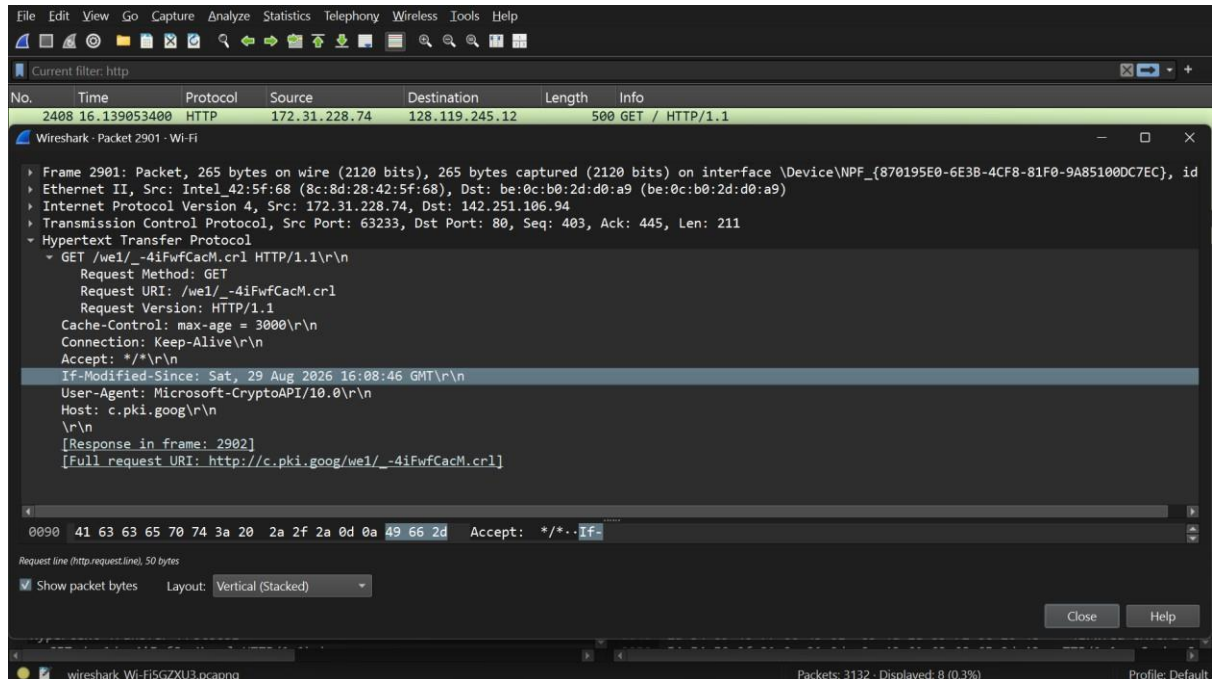
Yes, the server has returned the contents of the file. Extend line based text data: Congratulation you've download the file



19. Now inspect the contents of the second HTTP GET request from your browser to the server. Is there an “IF-MODIFIED-SINCE:” header line in the HTTP GET message? If so, what information follows the “IFMODIFIED-SINCE:” header line? (second link - yes)

ANS : 19

Yes. If-Modified-Since is present because the browser already has a copy of the page. It asks the server whether the page has changed since the last time it was downloaded.

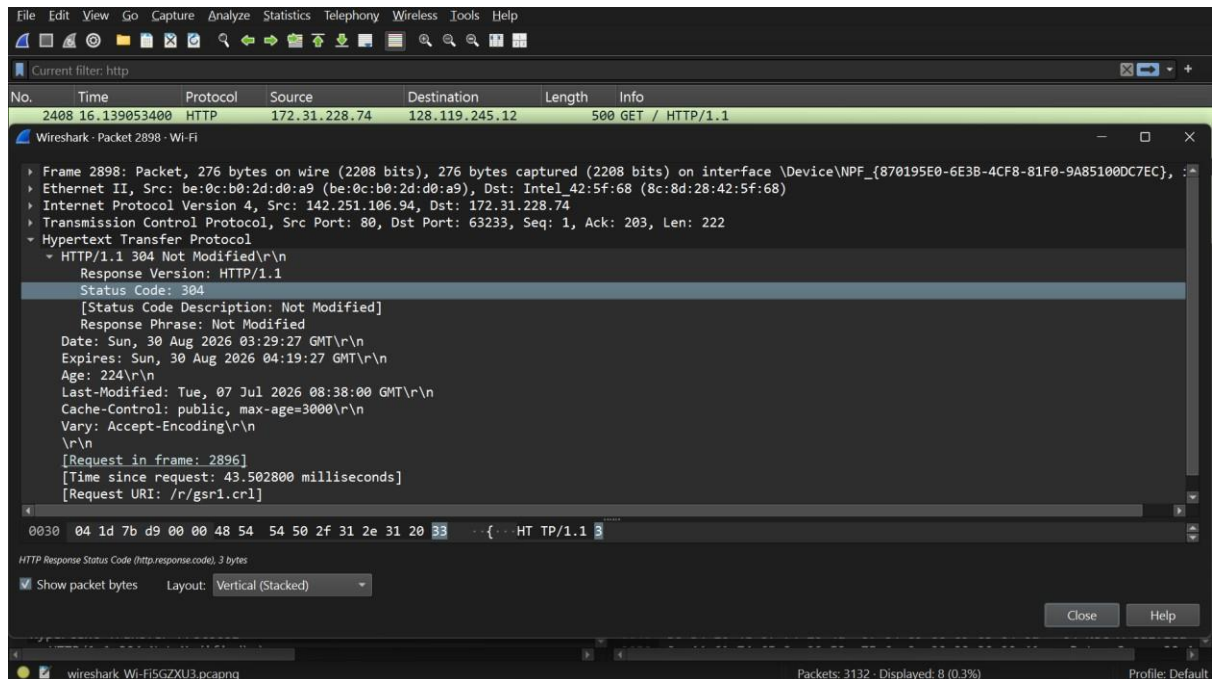


20.What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Has the server explicitly returned the contents of the file? Explain.

ANS :20

Status code:304 Response phrase: Not modified

No, the server has not returned the contents of the file as it is the response of the second request sent by the browser.



21. How many HTTP GET request messages has your browser sent?
Which packet in the trace contains the request for the Bill of Rights?

ANS : 21

Total HTTP GET request messages : 8

Wireshark packet capture showing HTTP GET requests. The packet list shows 8 GET requests. The packet details pane shows the structure of the first packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol.

No.	Time	Protocol	Source	Destination	Length	Info
2408	16.139053400	HTTP	172.31.228.74	128.119.245.12	500	GET / HTTP/1.1
2443	16.418900800	HTTP	128.119.245.12	172.31.228.74	619	HTTP/1.1 301 Moved Permanently (text/html)
2896	63.974606000	HTTP	172.31.228.74	142.251.106.94	256	GET /r/gsr1.crl HTTP/1.1
2898	64.018108800	HTTP	142.251.106.94	172.31.228.74	276	HTTP/1.1 304 Not Modified
2899	64.048865700	HTTP	172.31.228.74	142.251.106.94	254	GET /r/r4.crl HTTP/1.1
2900	64.095201400	HTTP	142.251.106.94	172.31.228.74	276	HTTP/1.1 304 Not Modified
2901	64.122399800	HTTP	172.31.228.74	142.251.106.94	265	GET /we1/_-4iFwfCacM.crl HTTP/1.1
2902	64.166201400	PKIX-CRL	142.251.106.94	172.31.228.74	1175	Certificate Revocation List

000 bits on interface \Device\NPF_{870195E0-6E3B-4CF8-81F0-9A85100DC7EC}, interface address: a9 (be:0c:b0:2d:d0:a9)

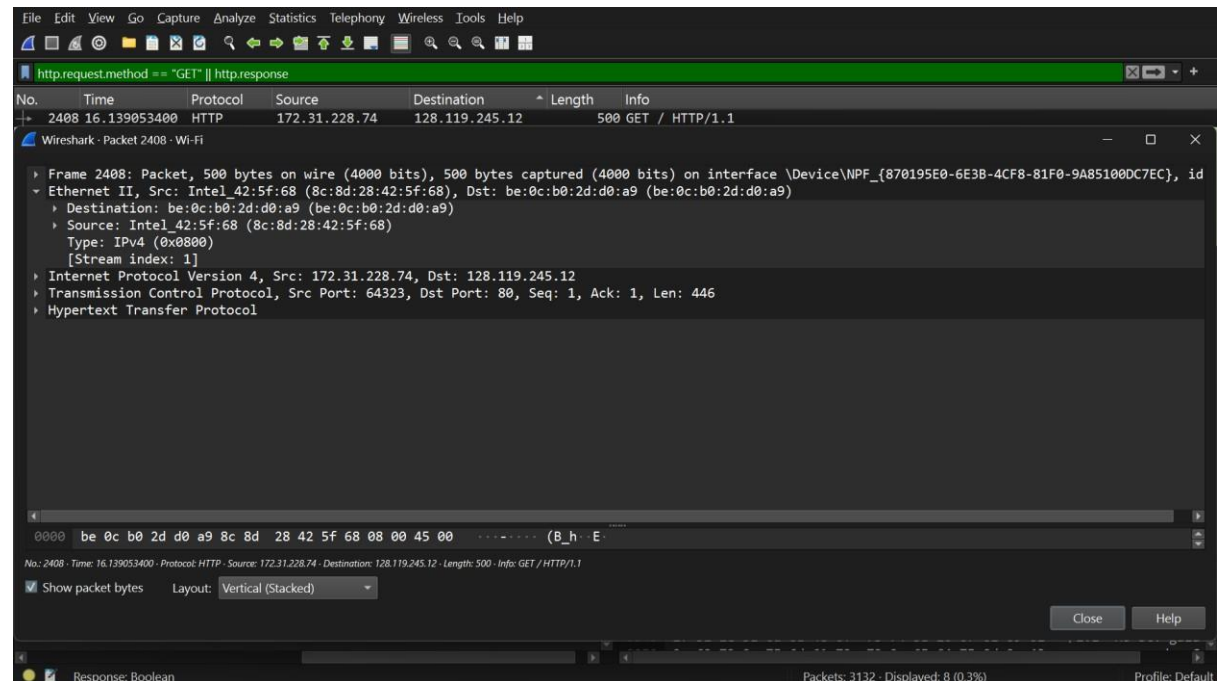
1, Len: 446

Response: Boolean

Packets: 3132 · Displayed: 8 (0.3%)

Profile: Defa

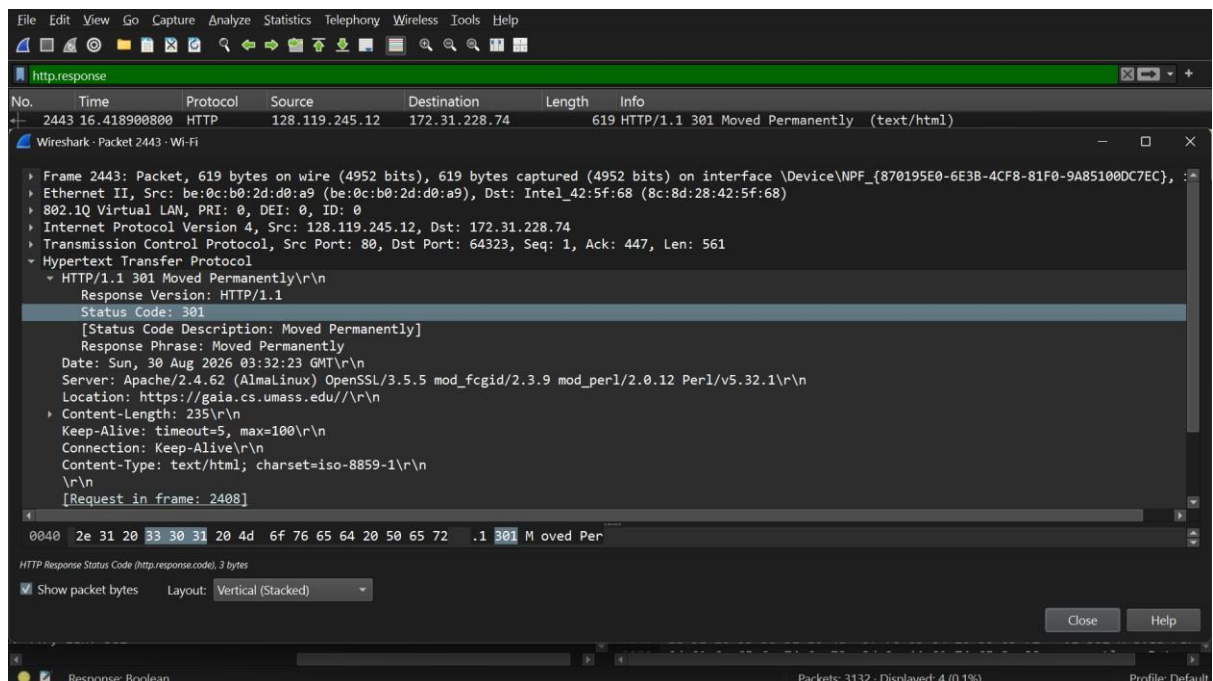
- Packet num is 2408 of get request



22.Which packet in the trace contains the status code and phrase associated with the response to the HTTP GET request? What is the status code and phrase in the response?

ANS : 22

Status Code : 301 Response Phrase : Moved Permanently Packet no of server response is : 2443



23. Discuss http GET and RESPONSE HTTP GET Request: An HTTP GET request is sent by the browser (client) to the server to request a web page or other resource. It contains information such as the requested URL, HTTP version, and request headers.

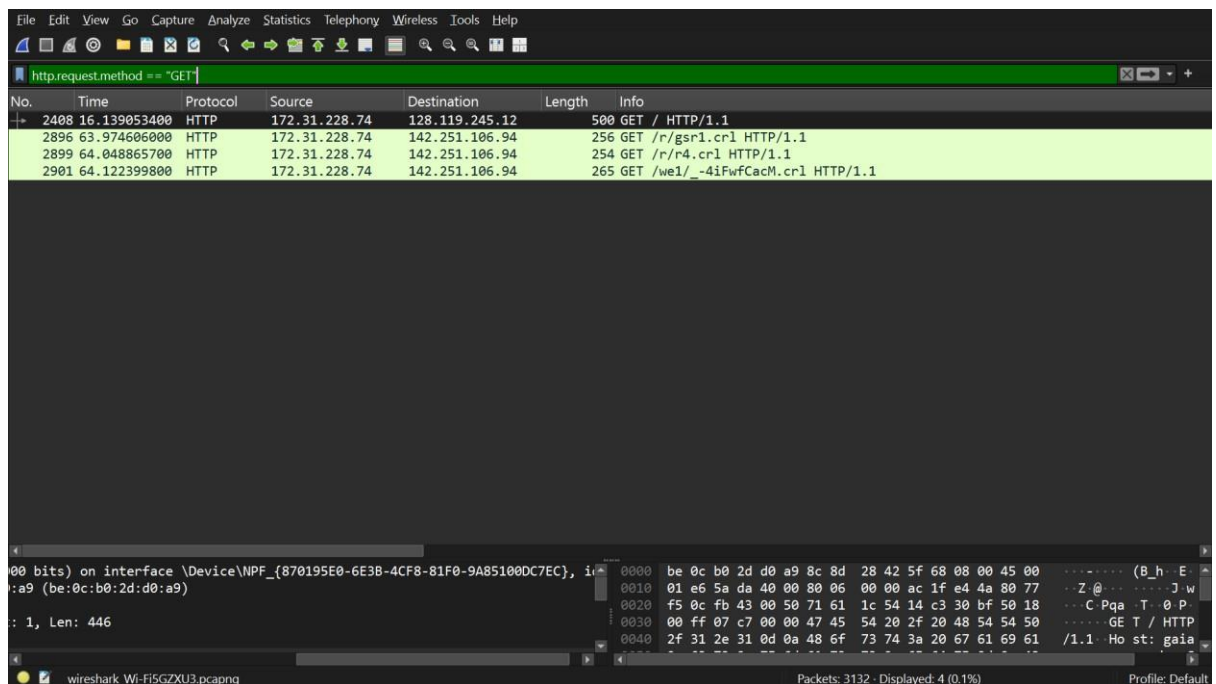
HTTP Response:

An HTTP Response is sent by the server back to the browser after receiving the GET request. It contains the status code (such as 200 OK or 304 Not Modified), response headers, and, if required, the requested web page or file.

24.Count the total number of HTTP GET requests.

ANS : 24

Displayed : 4



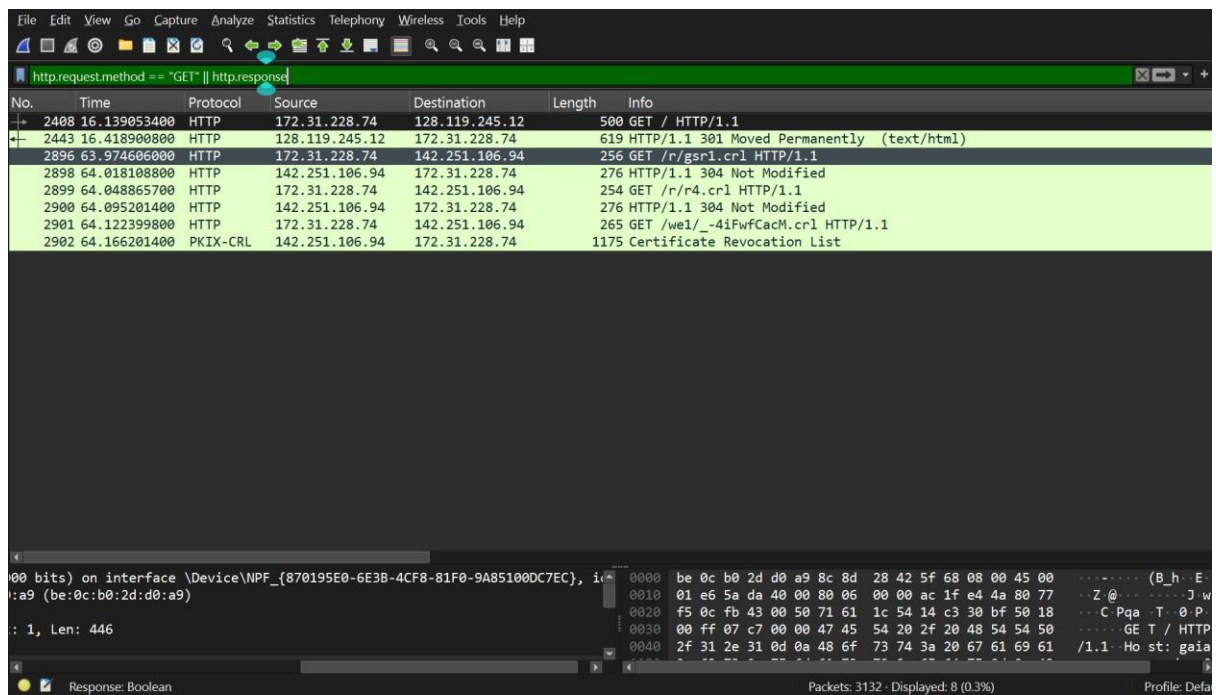
The image shows a Wireshark network traffic capture. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. A filter bar at the top displays the filter 'http.request.method == "GET"'. The main packet list table shows four filtered packets, all of which are HTTP GET requests. The bottom pane shows the details of the selected packet (No. 2901), including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol sections. The status bar at the bottom indicates 'Packets: 3132 · Displayed: 4 (0.1%)' and 'Profile: Default'.

No.	Time	Protocol	Source	Destination	Length	Info
2408	16.139053400	HTTP	172.31.228.74	128.119.245.12	500	GET / HTTP/1.1
2896	63.974606000	HTTP	172.31.228.74	142.251.106.94	256	GET /r/gsr1.cr1 HTTP/1.1
2899	64.048865700	HTTP	172.31.228.74	142.251.106.94	254	GET /r/r4.cr1 HTTP/1.1
2901	64.122399800	HTTP	172.31.228.74	142.251.106.94	265	GET /we1/_-4iFwfCacM.cr1 HTTP/1.1

25.How much time elapsed between the first HTTP GET request from client and the HTTP response (OK) from server?

ANS : 25

Response time is : 16.418900800 Request time is : 16.139053400
Elapsed time is : 0.2798474



Wireshark network traffic capture showing an HTTP 301 redirect and subsequent GET requests for CSS and JavaScript files. The filter is `http.request.method == "GET" || http.response`.

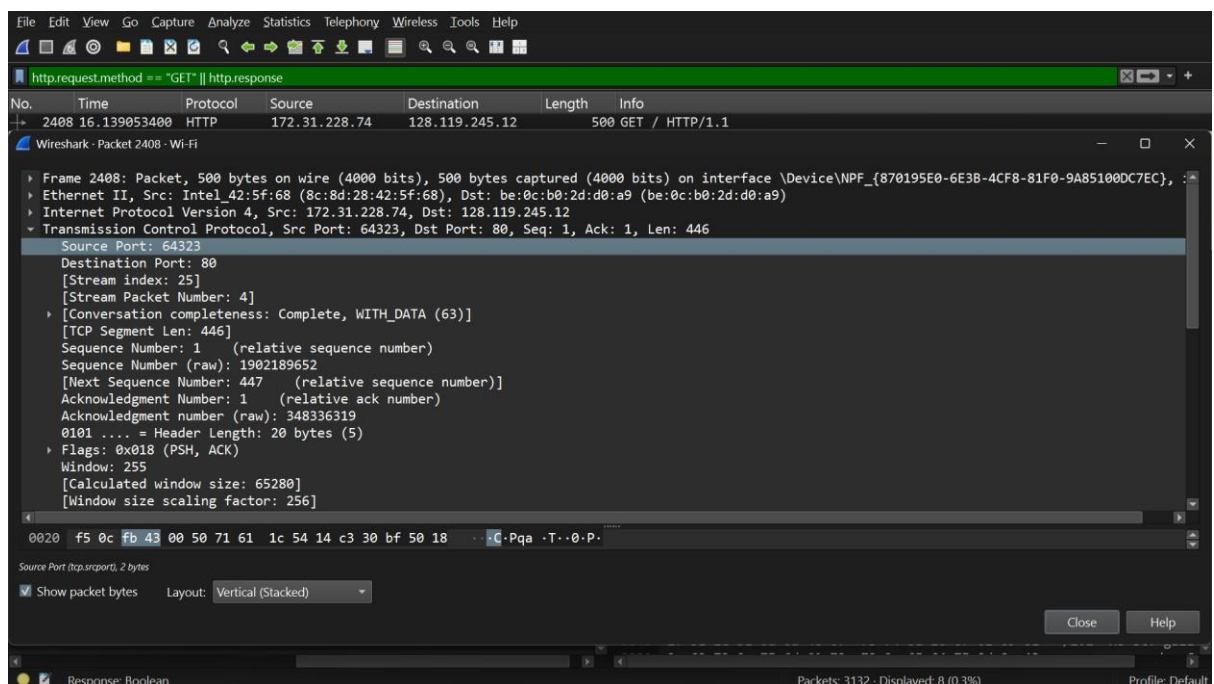
No.	Time	Protocol	Source	Destination	Length	Info
2408	16.139053400	HTTP	172.31.228.74	128.119.245.12	500	GET / HTTP/1.1
2443	16.418900800	HTTP	128.119.245.12	172.31.228.74	619	HTTP/1.1 301 Moved Permanently (text/html)
2896	63.974606000	HTTP	172.31.228.74	142.251.106.94	256	GET /r/gsr1.css HTTP/1.1
2898	64.018108800	HTTP	142.251.106.94	172.31.228.74	276	HTTP/1.1 304 Not Modified
2899	64.048865700	HTTP	172.31.228.74	142.251.106.94	254	GET /r/r4.css HTTP/1.1
2900	64.095201400	HTTP	142.251.106.94	172.31.228.74	276	HTTP/1.1 304 Not Modified
2901	64.122399800	HTTP	172.31.228.74	142.251.106.94	265	GET /we1/_-4iFwfCacM.css HTTP/1.1
2902	64.166201400	PKIX-CRL	142.251.106.94	172.31.228.74	1175	Certificate Revocation List

Packet 2902 details: 0000 bits on interface \Device\NPF_{870195E0-6E3B-4CF8-81F0-9A85100DC7EC}, interface address: a9 (be:0c:b0:2d:d0:a9). Ethernet II, Len: 446. Payload: Response: Boolean. Packets: 3132 · Displayed: 8 (0.3%) Profile: Defa

26. Identify the source and destination ports used to transfer the first HTTP GET request

ANS : 26

Source Port: 64323 Destination Port: 80

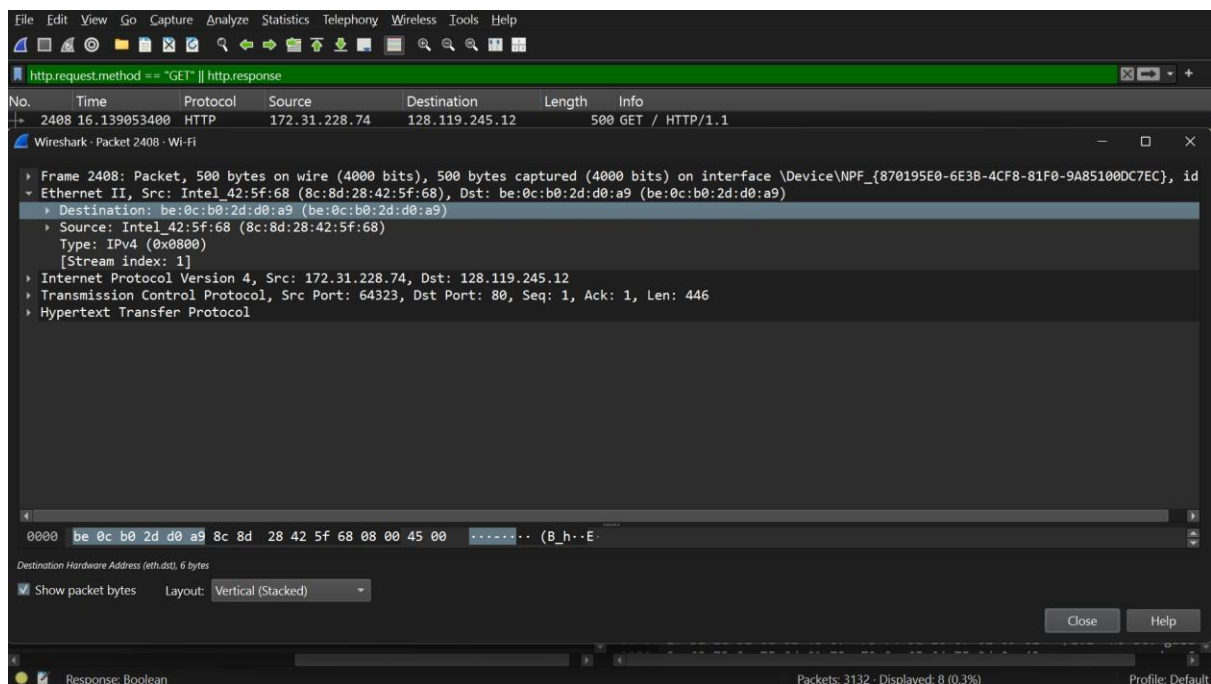


27. Identify the client and server ip address in the first HTTP GET request ?

ANS : 27

_Apply filter `HTTP.request.method == "GET"`, see the IP , within this you will find source ip address and destination ip addres.

Extend internet protocol



28. Identify the client and server MAC address in the first HTTP GET request?

ANS : 28

**Destination : be:0c:b0:2d:d0:a9 (be:0c:b0:2d:d0:a9) Source :
Intel_41:5f:68 (8c:8d:28:42:5f:68)**

